

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

Paper 2 Core Paper

9648/02

September 2016

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This document consists of **20** printed pages and **no** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 Fig 1.1 shows a molecule of tRNA and the enzyme that attaches the correct amino acid to it

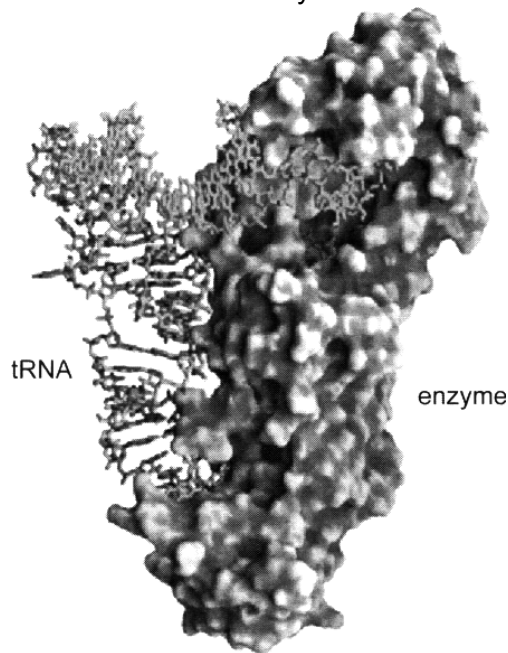


Fig. 1.1

- (a) With reference to **Fig. 1.1**, explain how the enzyme is suited to its function **[AJC 2011]**

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[3]

- (b) The average molecular weight of proteins encoded in the human genome is about 50,000 daltons. Given the average molecular mass of an amino acid is about 110 daltons and 5% of the initial transcript is converted to mature mRNA, estimate how long it will take a muscle cell to transcribe a gene for an average protein. Assume that the transcription rate is 20 nucleotides per second.

Show your working and express your answer to the nearest whole number (in minutes).

Answer: minutes [2]

- (c) Edeine is an antibiotic that inhibits protein synthesis but has no effect on either DNA synthesis or RNA synthesis. When added to the cell extract of an immature red blood cell, edeine stops protein synthesis after a short lag, as shown in Fig. 1.2. By contrast, cycloheximide, which is also an inhibitor, stops protein synthesis immediately. Protein synthesis is measured via radioactivity in haemoglobin.

Analysis of the edeine-inhibited cell extract showed that no polyribosomes remained at the time protein synthesis had stopped. Instead, all the globin mRNA were found associated with small ribosomal subunit and initiator tRNA.

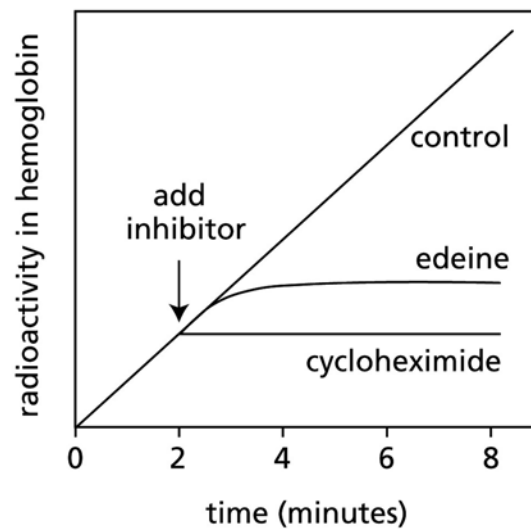


Figure 6-28 MBoc5: The Problems Book (© Garland Science 2008)

Fig. 1.2

- (i) Explain how protein synthesis is measured via radioactivity in haemoglobin.

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[1]

- (ii) Describe how edeine inhibits protein synthesis.

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[2]

- (iii) Explain why there is a lag between the addition of edeine and cessation of protein synthesis.

[2]

Protein synthesis in an organism is illustrated in Fig 1.3. (MJC 2011)

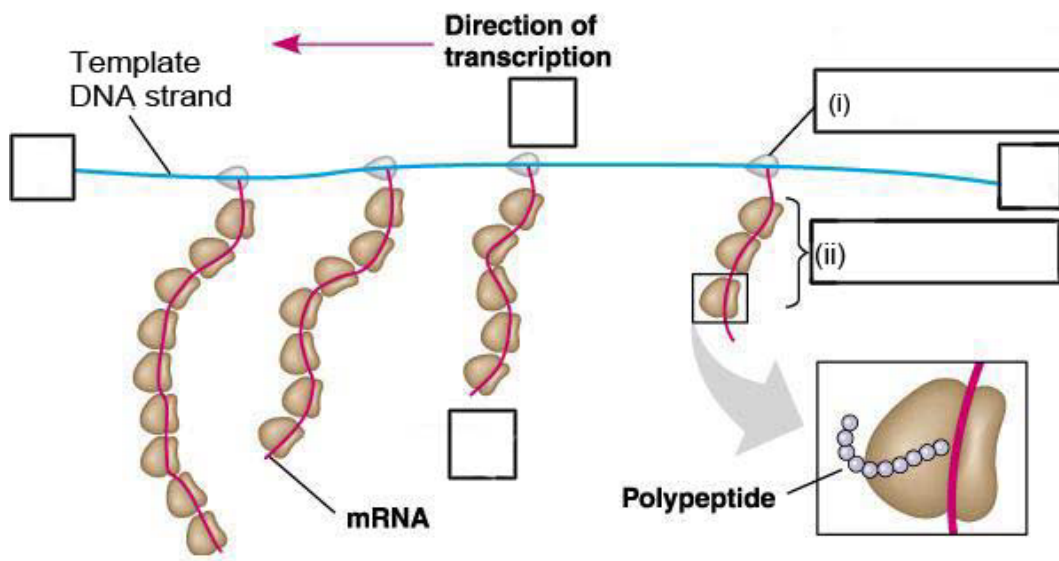


Fig. 1.3

- (a) Label the 5' and 3' ends of the mRNA and DNA template strand and structures (i) and (ii) on Figure 1.2. [2]

[Total: 12]

- 2 A gene for feather colour in chickens is carried on an autosome. This gene has two alleles, black (C^B) and splashed-white (C^W). When a male chicken with black feathers is mated with a female chicken with splashed-white feathers, all the offspring have blue feathers. This also occurs when a male chicken with splashed-white feathers is crossed with a female with black feathers.

Black
feathers



Splashed-white
feathers



Barred feathers



Another gene may cause stripes on feathers (barred feathers). The gene is carried on the X chromosome. The allele for barred feathers (X^A) is dominant to the allele for non-barred feathers (X^a). In chickens the male is homogametic, while the female is heterogametic.

- (a) A male chicken with black, non-barred feathers was crossed with a female chicken with splashed-white, barred feathers. All the offspring had blue feathers, but the males were barred and the females were non-barred.

- (i) Using the symbols given on the previous page draw a genetic diagram to show this cross.

Parental phenotype	Male, black, non-barred feathers	Female, splashed-white, barred feathers
Genotype		
Gametes		
Offspring genotypes		
Phenotypes		

[4]

- (ii) Explain how a farmer could use a breeding programme to find out the genotype of a male chicken with blue, barred feathers.

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[3]

- (b) The table below shows the genotypes and wool length in two breeds of sheep, X and Y, and their F1 hybrids.

	Genotype	Mean wool length/cm
Breed X	DDEEff	18
Breed Y	ddeeFF	14
F1 (breed X x Breed Y)	DdEeFf	16

When the F1 were crossed among themselves, the F2 offspring had a mean wool length ranging from 10cm to 22cm.

Assume that the inheritance of wool length in sheep depends upon alleles at three loci acting additively and that all variation in wool length in the F2 population is due to the segregation of alleles at these three loci.

How many different types of gametes would be produced by an organism of genotype DdEeFF, if all of the genes assort independently?

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[1]

What proportion of the F2 offspring are expected to have a mean wool length of 2 cm? Explain your reasoning.

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[4]

[Total: 12]

- 3 Phosphofructokinase (PFK) is a tetrameric enzyme that plays a central role in controlling the rate of glycolysis. In many organisms, the activity of PFK is enhanced allosterically by several substances, including ADP, and is inhibited allosterically by several other substances, including ATP and citrate. Fig. 3.1 shows part of the glycolysis reactions.

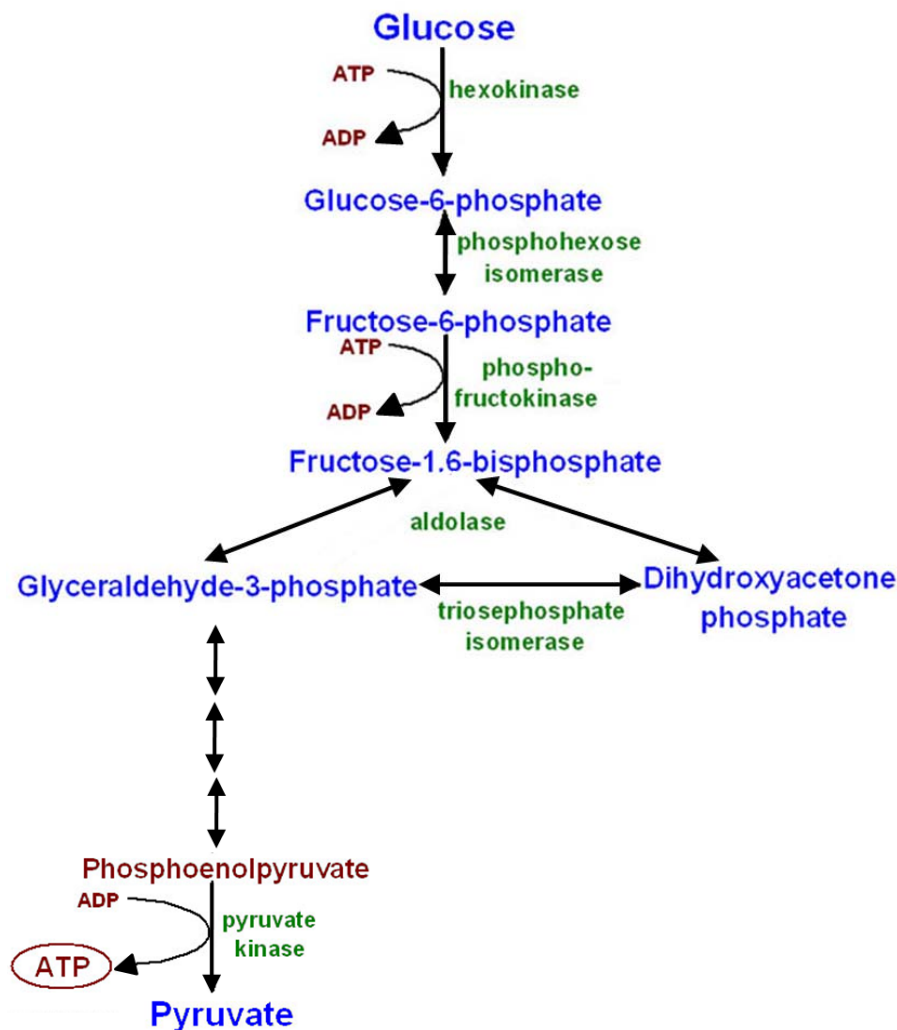


Fig. 3.1

- (a) Explain why phosphofructokinase is a quaternary protein. MJC 2011

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[1]

- (b) Besides PFK, hexokinase and pyruvate kinase also play a role in determining the overall rate of glycolysis.

With reference to Fig. 3.1, suggest why reactions catalyzed by these three enzymes determine the rate of glycolysis.

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[1]

- (c) Explain the importance of inhibition of PFK by citrate when citrate concentration in the cell is high.

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[2]

- (d) Describe how citrate inhibits PFK.

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[3]

- (e) In mammals, under anaerobic condition, both Krebs cycle and oxidative phosphorylation do not occur. Instead, lactate fermentation occurs.

Explain the significance of lactate fermentation.

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[2]

- (f) Hexokinase, the enzyme involved in the first step of glycolysis, is able to utilize various hexoses such as glucose, fructose and mannose.

Suggest how hexokinase is able to utilize different substrates.

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[2]

[Total: 12]

- 4 Norepinephrine is a stress hormone secreted during fight-or-flight response. Fig. 6.1 shows an example of a G-protein signalling pathway.

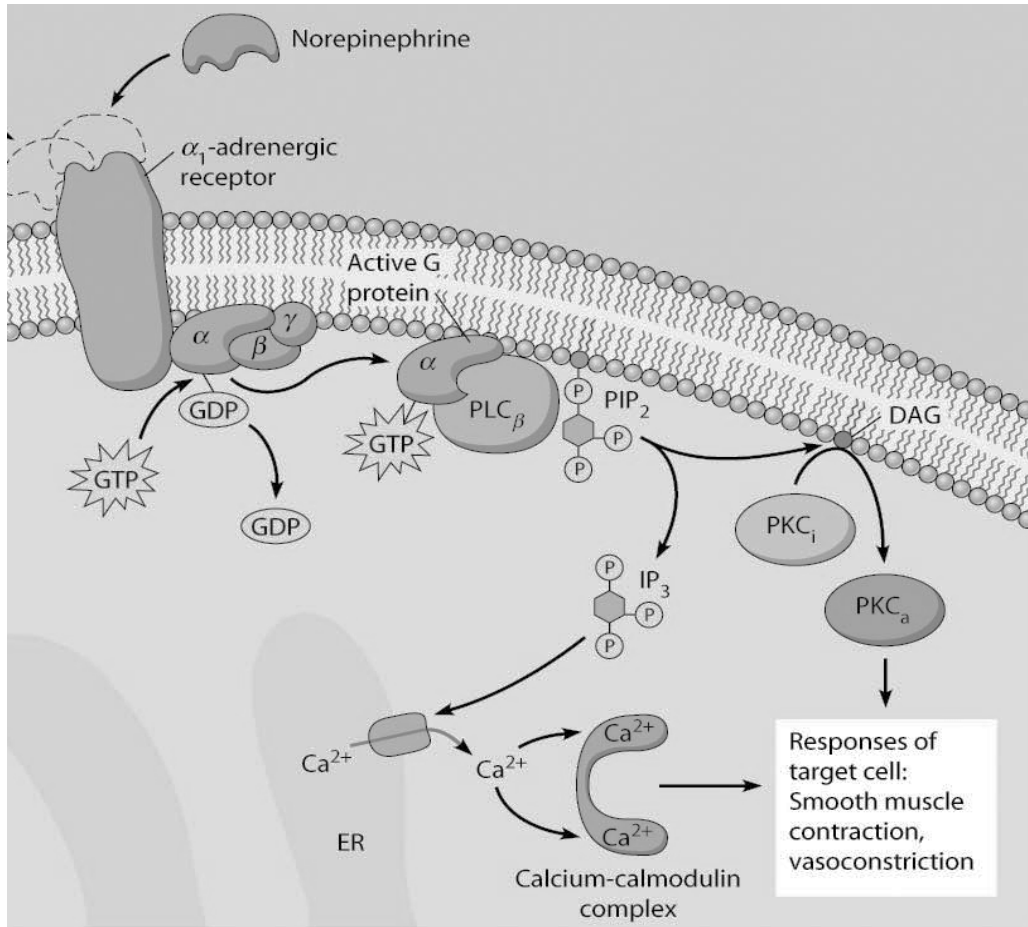


Fig. 4.1

- (a) With reference to Fig. 6.1, describe explain how the presence of norepinephrine leads to the opening of IP₃-gated Ca²⁺ channels.

[4]

- (b) With reference to Fig. 6.1, explain the significance of IP_3 and Ca^{2+} in the cell signalling pathway.

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[2]

- (c) The same signalling molecule norepinephrine usually targets more than one cell type. For e.g. muscle, heart and liver cells.

Explain how the same norepinephrine molecule can have different effects on different cells.

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[2]

- (d) Suggest how signal termination can occur.

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[1]

Most people with type I diabetes inject insulin. A recent product contains insulin that can be administered using a nasal spray. The spray is inhaled and the insulin is taken up through the lungs.

Fig. 6.2 shows the concentration of insulin in the blood plasma in the 480 minutes after injecting or inhaling insulin. In both cases, the insulin was of the same type, obtained from genetically engineered *Escherichia coli*.

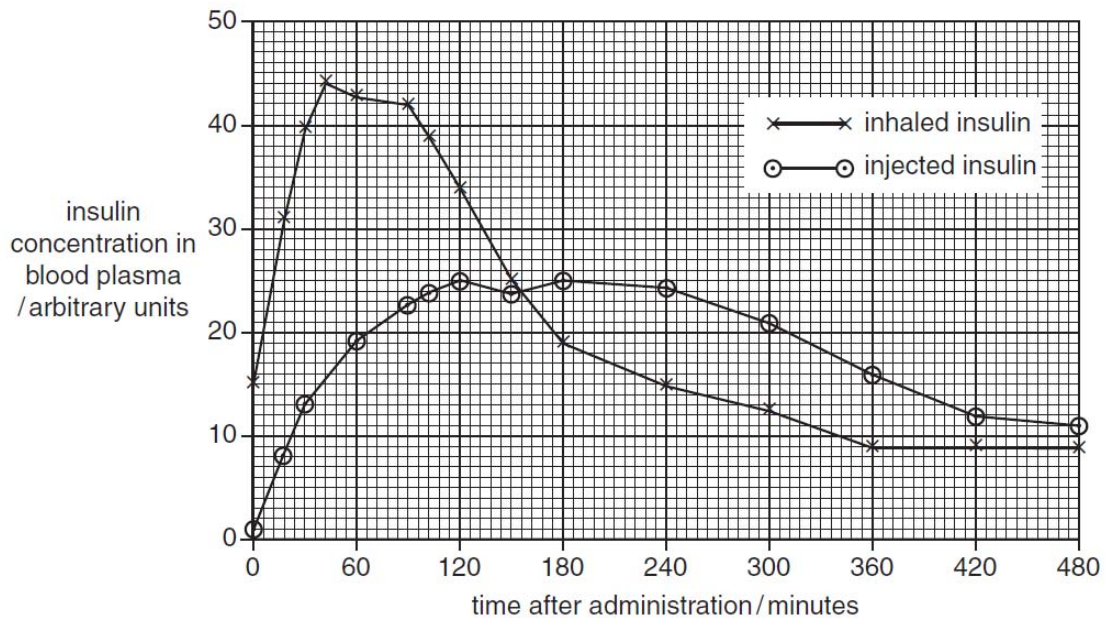


Fig 6.2

Fig. 6.3 shows the concentration of glucose in the blood plasma in the 480 minutes after injecting or inhaling insulin.

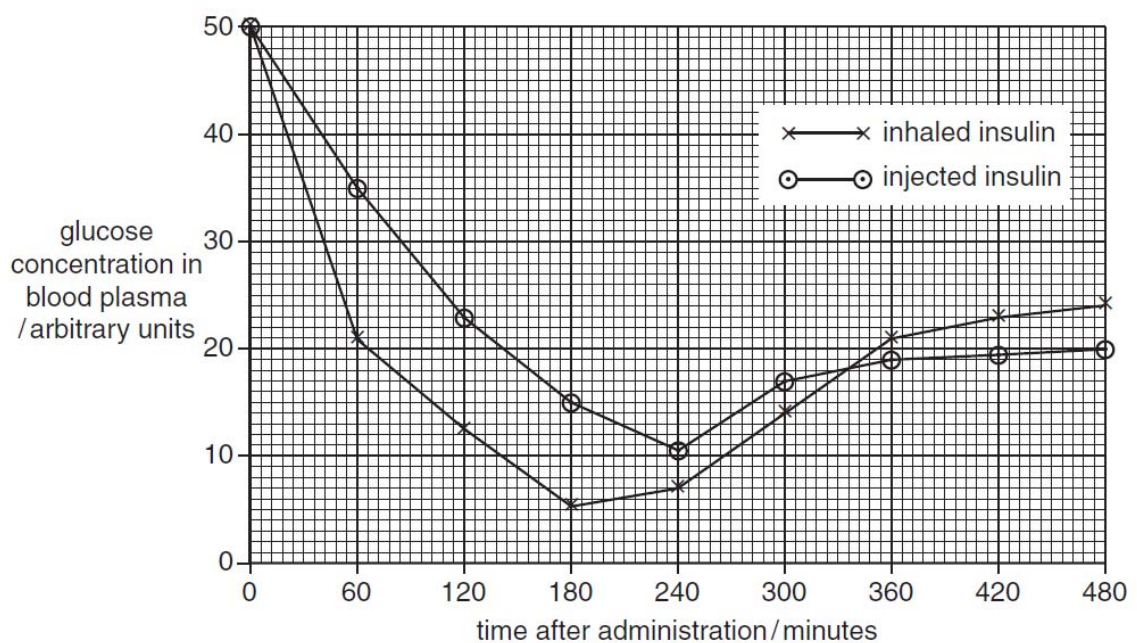


Fig 6.3

- (e) Compare the results for injected insulin and inhaled insulin shown in Fig. 6.2.

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[2]

- (f) With reference to Fig. 6.2, explain the differences in the blood glucose levels after injecting or inhaling insulin shown in Fig. 6.3.

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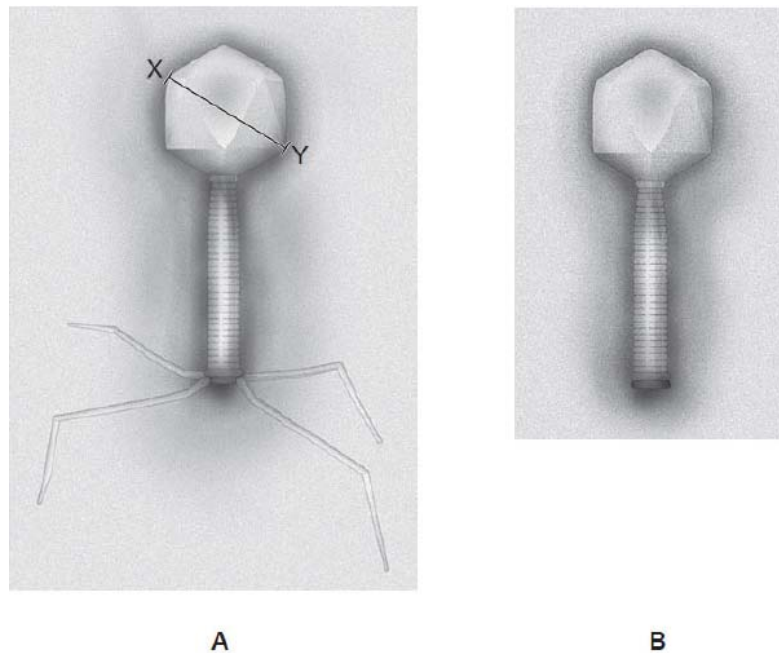
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[3]

[Total: 14]

- 5 Lambda is a bacteriophage that uses *Escherichia coli* as its host cell. Fig. 5.1 A is an electron micrograph (EM) of a wild-type bacteriophage lambda while Fig. 5.1 B is an EM of a laboratory-cultured lambda. **MJC 2010**



- (a) State one main difference between the two bacteriophages.

Presence of tail fibres in phage A

[1]

- (b) Studies have shown that laboratory-cultured bacteriophage can bind to the host cell, *Escherichia coli*. But no new phages are formed.

Suggest how the laboratory-cultured bacteriophage binds to the host cell, *Escherichia coli*.

Binding to host cell via base plate

[1]

- (c) Suggest why no new phages were formed.

Phage DNA integrates into bacterial genome as prophage ; undergoing lysogenic cycle

[1]

- (d) Explain how lambda phage enters the host cell.

Lambda phage recognises and binds / attaches to specific receptor sites on host cells' surface (attachment phase)

Penetrates into host by injecting its DNA into the cytosol of the host bacteria

[2]

- (e) Suggest why it is sometimes difficult to distinguish between a plasmid and the DNA of the bacteriophage lambda inside an *Escherichia coli*, when viewed under the electron microscope.

Both may be similar in size / circular in shape

[1]

Some of the *Escherichia coli* are infected with the laboratory-cultured lambda phage. The bacteria are initially cultured in a nutrient medium without X-gal. The bacteria colonies produced are replica plated onto two agar plates, one containing X-gal and lactose and the other containing X-gal without lactose.

There is no glucose in either plates. Fig. 5.2. shows the results of this experiment.

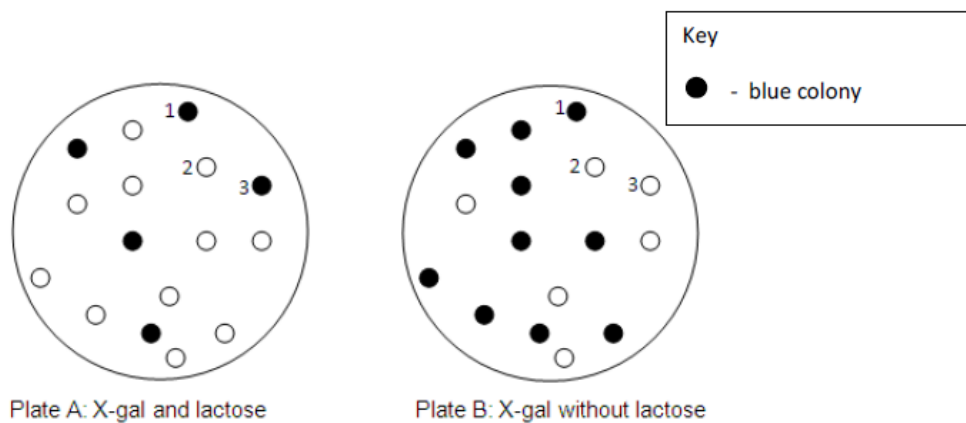


Fig. 5.2

- (f) Account for the observations seen in:

- (i) Colony 1

Colony 1 is blue in both plates because transcription of Lac Z gene is turned on all the time so β galactosidase is continuously translated to break down X-gal into a blue compound

Phage DNA is integrated into the operator by transduction and repressor cannot bind to operator OR phage DNA is integrated into the lacI gene and no repressor is produced;

[2]

(ii) Colony 2

Colony 2 is white in both plates because transcription of lac Z gene cannot take place and β galactosidase is not translated ;

Viral DNA is integrated into the promoter and RNA polymerase cannot bind to promoter OR viral DNA is integrated into lac Z gene and lac Z gene is disrupted leading to insertional inactivation ;

[2]

(iii) Colony 3

Observation for colony 3 is as expected. Blue when lactose is present and white when lactose is absent / regulation of lac operon is normal ;

Viral genome is not integrated into the lac operon / no transduction occurs ;

[1]

[Total: 11]

6 (a) With reference to the *lac* operon, explain what is meant by : CJC 2011 Q3

(i) regulatory genes;

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[2]

(ii) structural genes

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[2]

(b) Fig. 6.1 below is an electron micrograph of a microorganism that is isolated from the stools of cholera patients.



Fig. 6.1

With reference to Fig. 6.1, state **two** structural differences between the chromosome of this type of microorganism and a human cell.

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[2]

- (c) The enzyme lactase is responsible for lactose metabolism in humans and is encoded by a gene on chromosome 2.

Based on your understanding of eukaryotic and prokaryotic genomes, describe how the genes in humans and *Escherichia coli* that are responsible for lactose metabolism differ in terms of:

- (i) their organisation;

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[1]

- (ii) the control of gene expression.

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[2]

- (d) Streptomycin is an antibiotic that binds to bacterial ribosomes.

- (i) Explain how this effect of streptomycin prevents the growth of bacteria.

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[2]

- (ii) Suggest why streptomycin does not affect the growth of human cells.

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[1]

[Total: 12]

- 7 **Fig. 7.1** shows the distribution of the giant tortoises across the various islands of the Galápagos archipelago. Some sub-species are distributed across the southern part of the largest island named Isabela. In addition, three other species, *G. chilensis*, *G. carbonaria*, and *G. denticulate* are also found on the west coast of Ecuador, a country in the South America continent.

The giant tortoises had been previously classified based on morphological differences, primarily using carapace (shell) shape, which varies from domed to saddleback with intermediate forms also occurring. [AJC 2011]

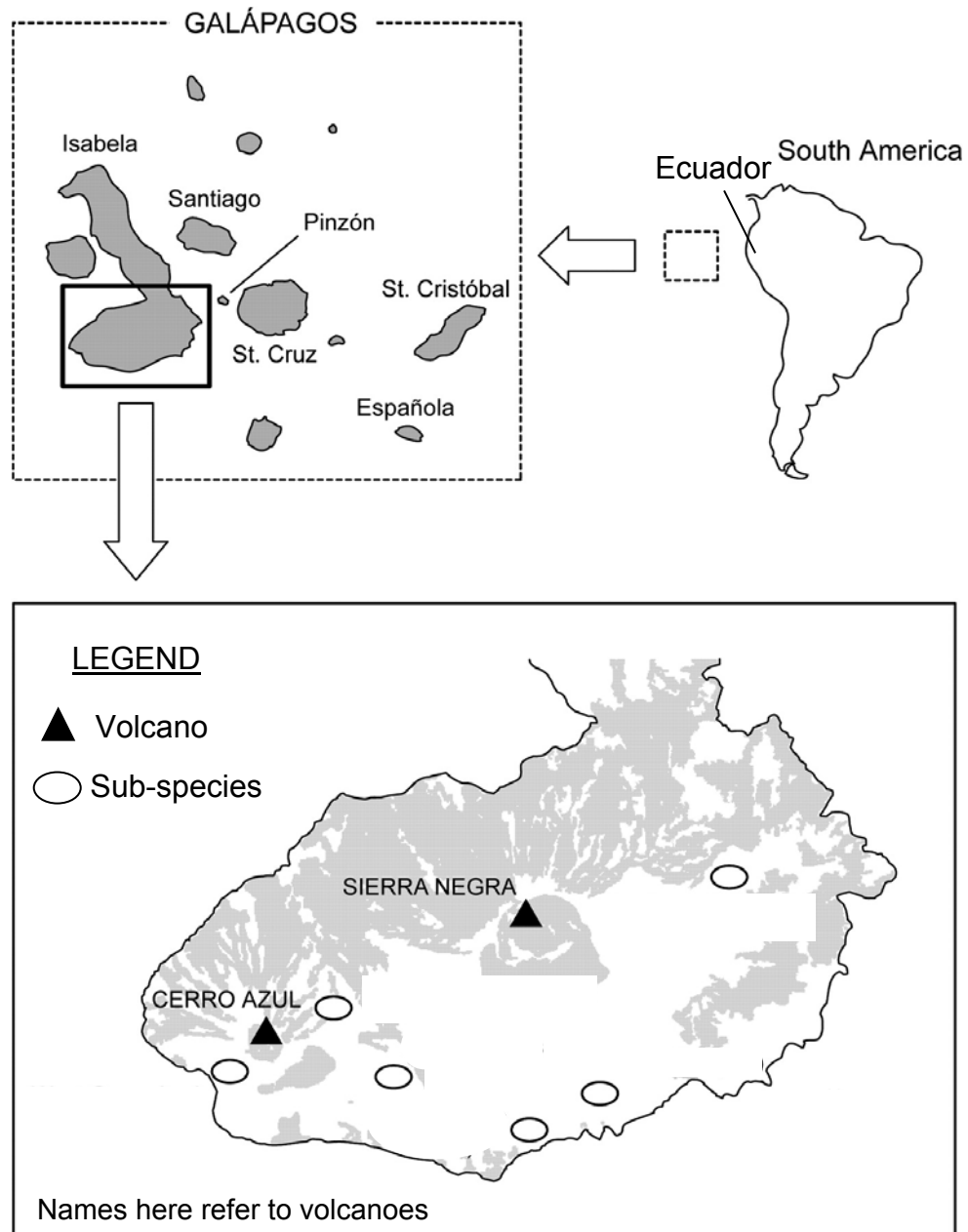


Fig. 7.1

- (a) With reference to Fig. 7.1, explain how biogeography supports the evolutionary deductions that the tortoises are all evolutionally related.

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[2]

- (b) Suggest why the previous classification using carapace shape may not be an accurate way of determining the evolutionary relationships accurately.

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[2]

- (c) Suggest why the sub-species found on the southern part of Isabela island did not evolve to become distinct species.

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[2]

In another separate experiment, scientist were investigating on the Red Queen hypothesis which state that sexual reproduction persists because it enables many species to rapidly evolve new genetic defenses against parasites that attempt to live off them.

Scientists have tested this idea by observing different groups of small fish *Poeciliopsis* species (Gila topminnow) in Mexico. Some populations of the topminnow reproduce sexually, while others practice parthenogenesis. Parthenogenesis occurs when females produce offspring without any male contribution and the female's gametes develop directly into female offspring.

Topminnows are constantly parasitized by 'black spot disease' caused by black spot worms that encyst in the skin. Parasitized topminnows rarely survive. The researchers found that identical populations of the asexually reproducing topminnows harbored many more black-spot worms than did those producing sexually, a finding that fit the Red Queen hypothesis: the sexual topminnows could devise new defenses faster by recombination than the asexually producing clones.

However, it was observed that after a drought, the sexually reproducing topminnows were more heavily parasitized than the cloned topminnows. This process of whereby chance events cause the allele frequency to change unpredictably is known as genetic drift.

(i) Name the evolutionary event that resulted in this genetic drift.

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[1]

(ii) Explain why there was an increase in the number of parasitized sexually reproducing topminnows after the drought.

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[2]

The Neutral Theory of Molecular Evolution proposed that frequencies of alleles are not affected by natural selection but may increase or decrease as a result of genetic drift.

(iii) Briefly describe the Neutral Theory of Molecular Evolution.

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[3]

[Total: 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a) Describe the changes involving homologous chromosomes during meiosis, and explain how these changes ensure genetic variation and genetic stability. [10]
- (b) Describe the development of cancer as a multi-step process. [6]
- (c) Describe the structure and functions of the centrioles in the cell. [4]
- 9 (a) Explain the role of control elements in regulating gene expression in eukaryotes. [8]
- (b) Using named examples, describe the process and significance of gene amplification in eukaryotes. [8]
- (c) Explain why the ability to rapidly degrade mRNA can be an adaptive advantage for prokaryotes. [4]